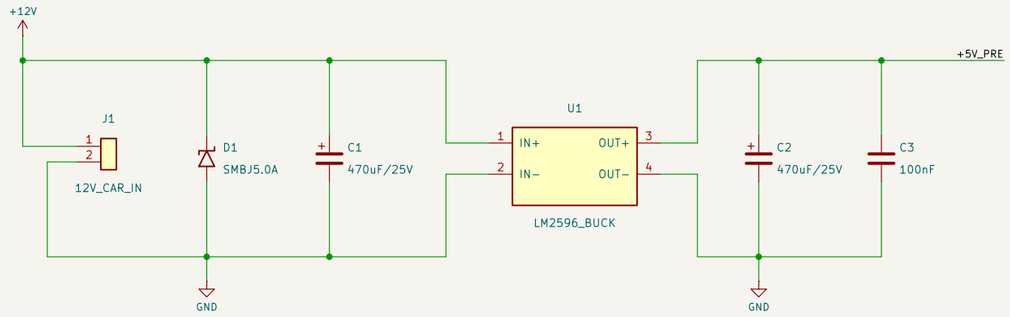
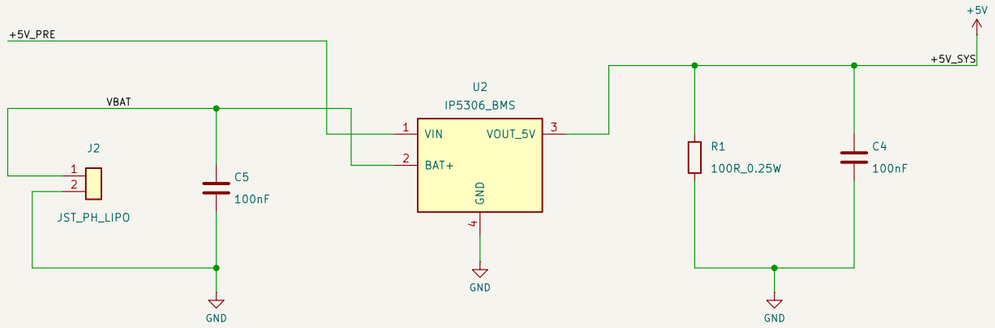
SECTION 1: AUTOMOTIVE 12V–24V INPUT & LM2596 BUCK PRE-REGULATOR

Pre-regulates 12V automotive power to +5V_PRE (2A capacity)



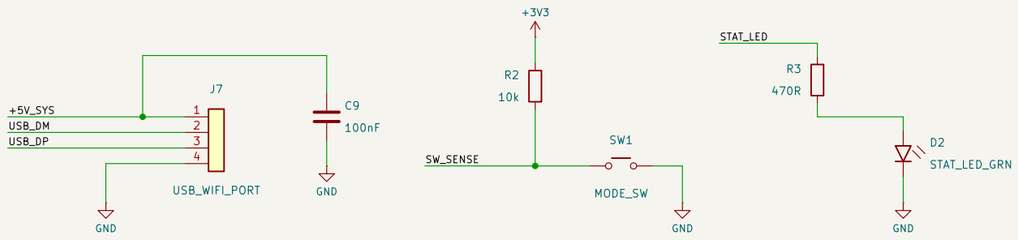
SECTION 2: IP5306 BMS & 3.7V LIPO BATTERY BUFFER

Seamless UPS power path: Charges 1000mAh LiPo & boosts to +5V_SYS



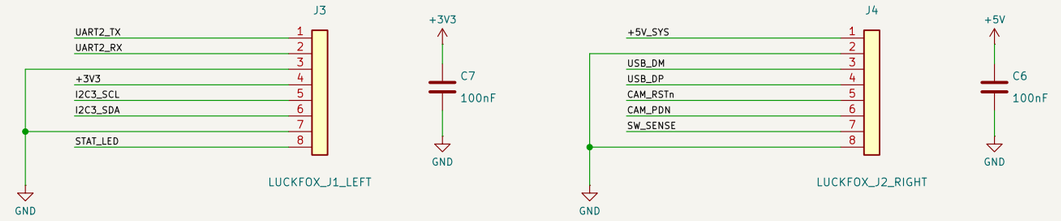
SECTION 4: USB 2.0 WI-FI DONGLE PORT & USER CONTROLS

Host port for RTL8188EUS Wi-Fi dongle, Mode Tactile Switch & Status LED



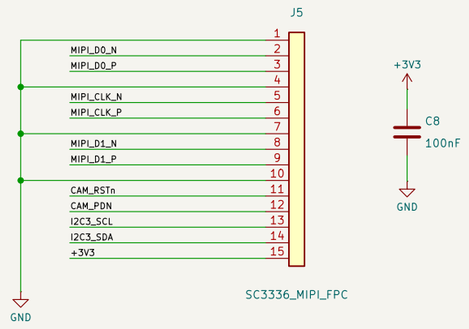
SECTION 3: LUCKFOX PICO MINI B (RV1103) SBC EXPANSION SOCKETS

Dual 8-Pin 2.54mm pitch female header sockets (J3 Left / J4 Right)



SECTION 5: SC3336 CAMERA MIPI CSI-2

15-Pin 0.5mm FPC (2-Lane MIPI + I2C)



DASHCAM PROTOTYPE ARCHITECTURE SUMMARY

- SoC: Rockchip RV1103 (Cortex-A7 @ 1.2GHz + 0.5TOPS NPU)
- Sensor: SmartSens SC3336 3MP CMOS (2304x1296 @ 30fps)
- Video: Hardware H.264/H.265 encoder via RTSP streaming
- Power Input: 12V–24V Car Accessory Socket via LM2596 Buck
- UPS Buffer: IP5306 + 3.7V 1000mAh LiPo (Safe shutdown)
- Connectivity: RTL8188EUS 2.4GHz 802.11b/g/n USB Wi-Fi
- PCB Spec: 2-Layer FR4, 1.6mm thickness, 1oz copper
- Form Factor: 65.0 x 42.0 mm (Matching Luckfox Pico Mini B)

Status: RELEASED
Luckfox Pico Mini B (RV1103) + SC3336 + IP5306 + RTL8188EUS
Discipline: EMBEDDED SYSTEMS & HARDWARE
Engineer: Souvik H | Regn: IN-ENG-2022-22824
DASHCAM PROTOTYPE PROJECT

Sheet: /
File: carrier_board.kicad_sch

Title: DASHCAM CARRIER BOARD SCHEMATIC

Size: A3	Date: 2026-10-11	**Rev: 1.4**
KiCad E.D.A. 10.0.7		Id: 1/1